MODULE ASSIGNMENT PYTHON 5

MARKETING CAMPAIGN PERFORMANCE INSIGHTS

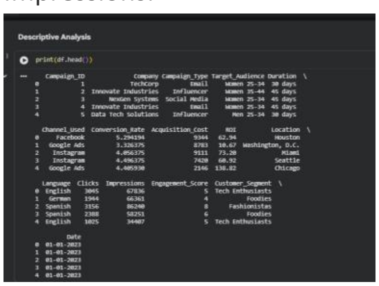
Executive Summary

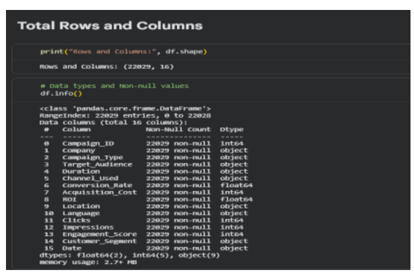
In the digital marketing environment, continuous evaluation of campaign performance is mandatory to maximize return on investment (ROI), minimize acquisition costs, and boost conversion rates. This report provides a structured, non-technical analysis of marketing campaign performance across diverse channels, audience segments, campaign types, and geographical regions.

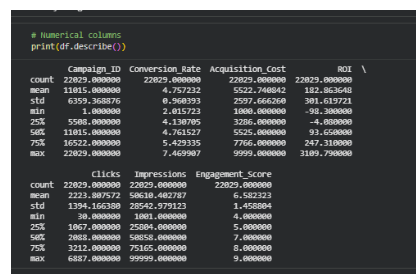
Descriptive Analysis & Dataset Overview

Dataset Structure & Integrity

- **Data Profiling:** The analysis evaluates dataset shape, column formats, memory consumption, and non-null values to ensure seamless analytical processing.
- **Numerical Summary:** Descriptive statistics (Mean, Standard Deviation, Median, Min, and Max) are computed for key numerical metrics including ROI, Acquisition Cost, Conversion Rate, Clicks, and Impressions.







Categorical Data Exploration

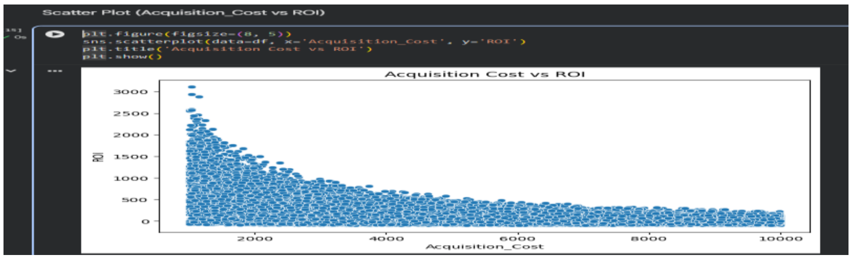
- **Campaign Identification:** Distinct count of Campaign IDs is recorded to confirm individual campaign tracking integrity.
- **Category Frequencies:** Identifies unique geographical locations and customer segment profiles while summarizing distribution counts across campaign types and promotion channels.

Exploratory Data Analysis & Strategic Insights

Campaign Performance Dynamics

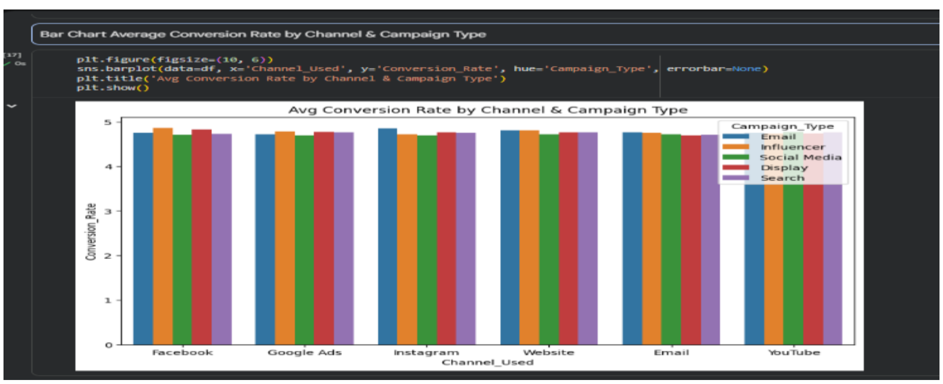
Scatter Plot (Acquisition Cost vs. ROI)

Examines the relationship between customer acquisition expenditure and overall campaign profitability to evaluate spending efficiency.



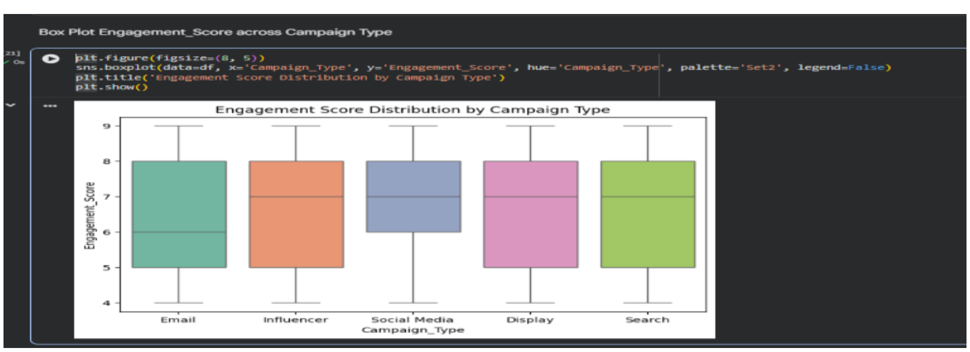
BAR CHART (Channel & Type Conversion Rate)

Identifies which marketing channels (e.g., Google Ads, Social Media) yield the highest average conversion rate when paired with specific campaign types.



BOX PLOT (Engagement Score Distribution)

Analyzes customer interaction scores (1 to 10) across campaign categories to identify formats that consistently drive higher user interest.

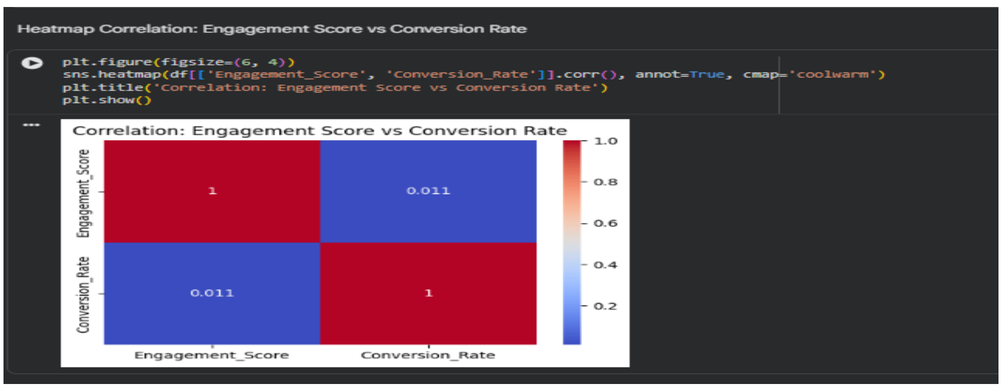


BAR CHART (Company Profitability Comparison)

Compares average ROI across operating companies/brands to benchmark relative marketing success.

HEATMAP (Engagement vs. Conversion Correlation)

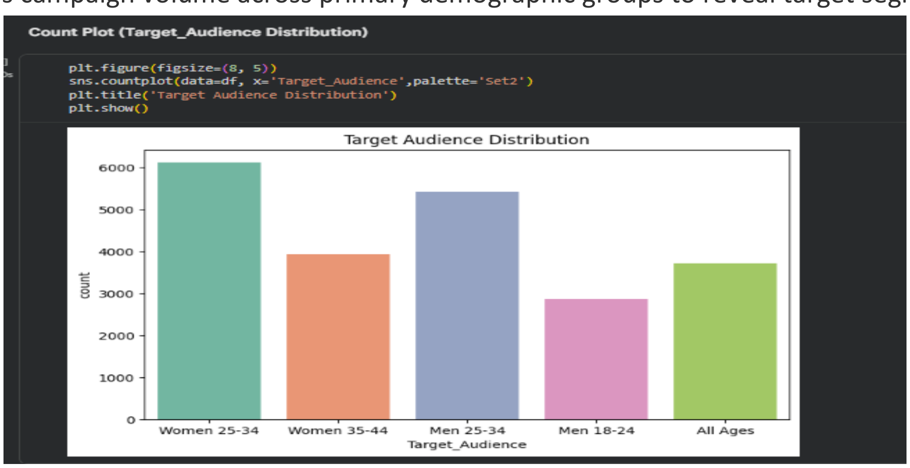
Measures how strongly user engagement score correlates with actual conversion rates.



Customer Segmentation & Localization

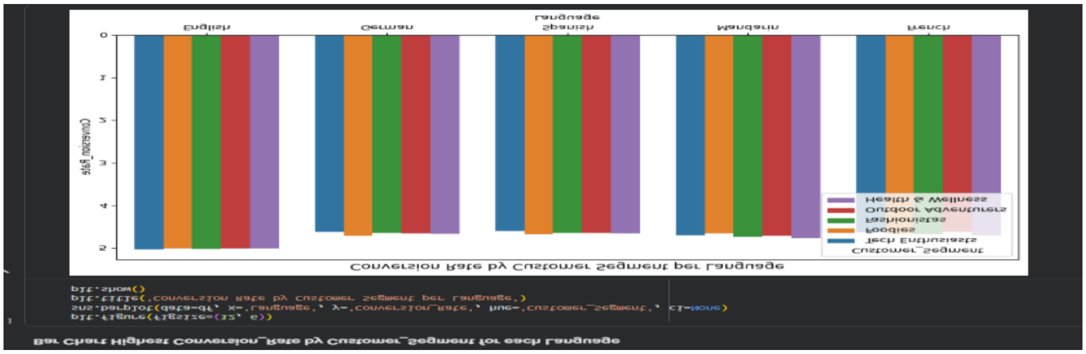
COUNT PLOT (Target Audience Reach)

Assesses campaign volume across primary demographic groups to reveal target segment focus.



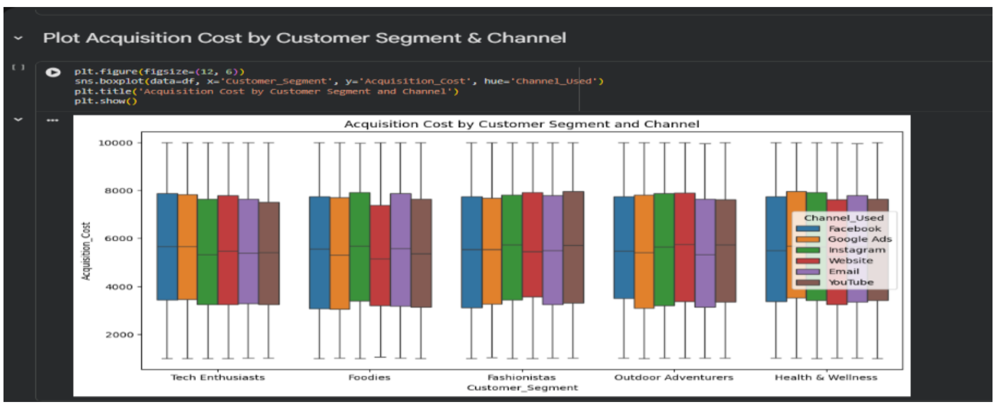
BAR CHART (Segment Efficiency by Language)

Pinpoints customer segments achieving peak conversion rates across distinct content languages.



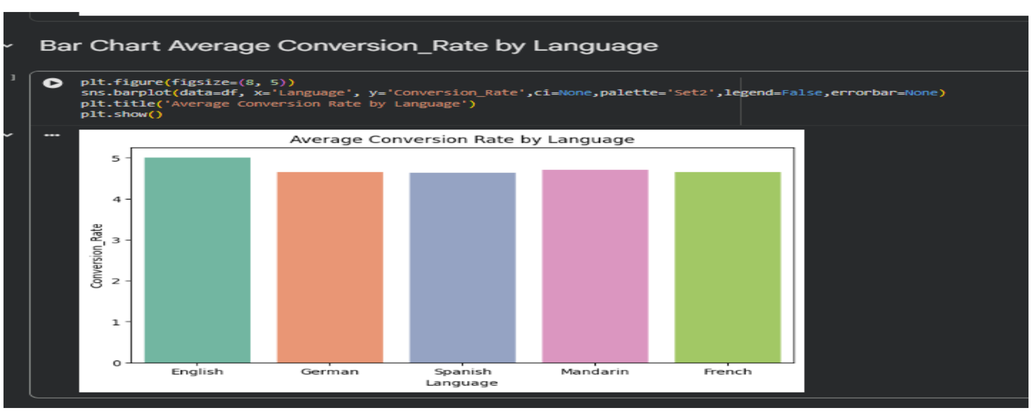
PLOT CHART (Acquisition Expense Spread)

Evaluates variations in customer acquisition expenses per segment across selected marketing channels.



BAR CHART (Language Effectiveness)

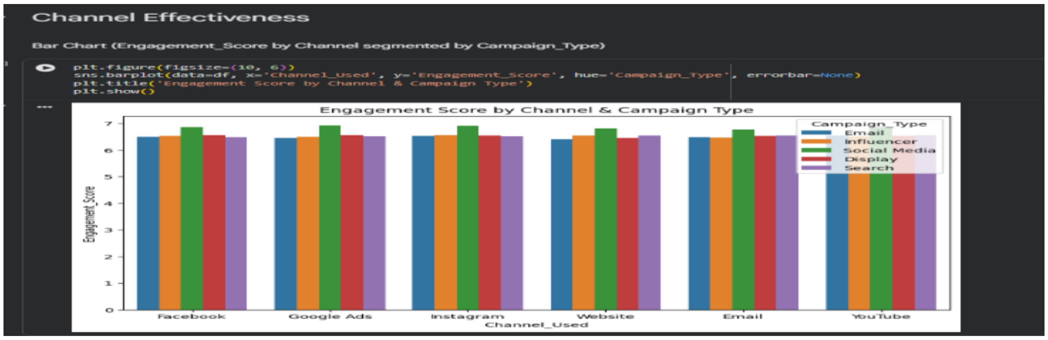
Compares aggregate conversion rates by language to optimize localized ad messaging.



Channel Effectiveness & Attribution

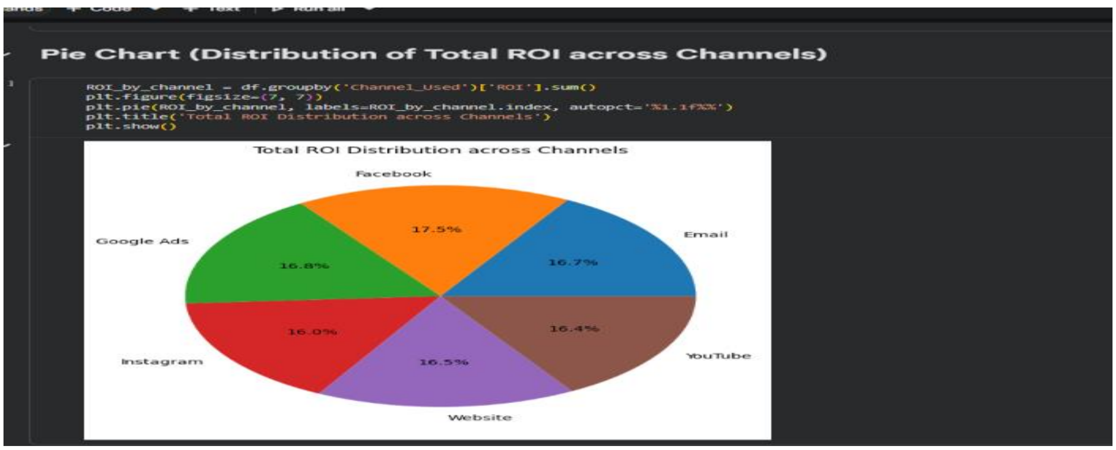
BARCHART (Channel Engagement Levels)

Highlights top-performing interaction platforms across email, display, social media, and search campaigns.



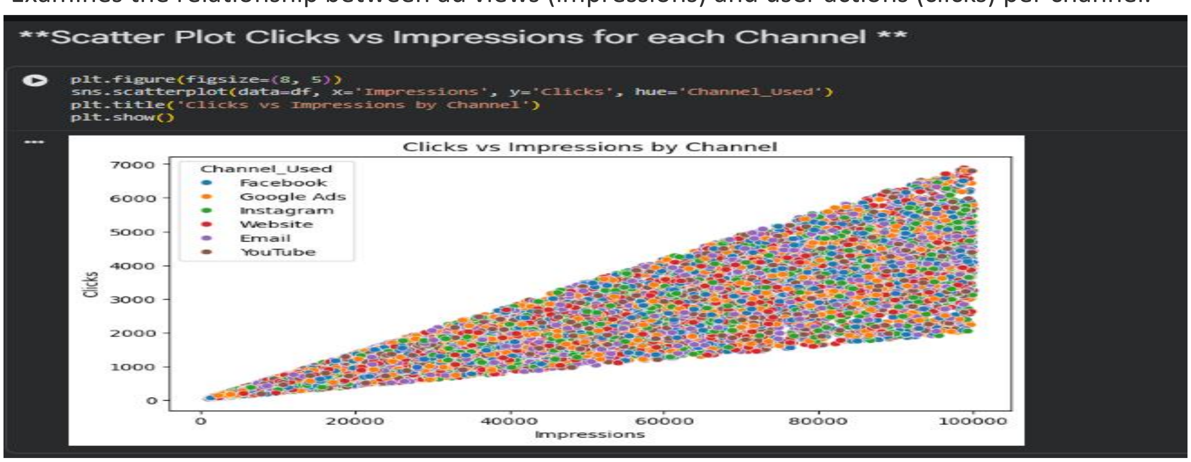
PIE CHART (Cumulative ROI Share)

Illustrates the percentage share of overall profitability generated by each promotion medium.



SCATTER PLOT CHART (Clicks vs. Impressions (CTR Potential)

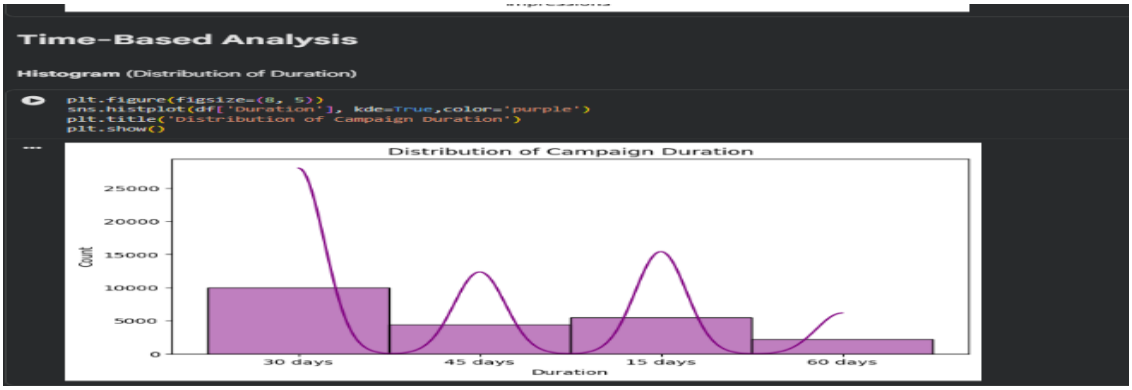
Examines the relationship between ad views (impressions) and user actions (clicks) per channel.



Time-Based Performance

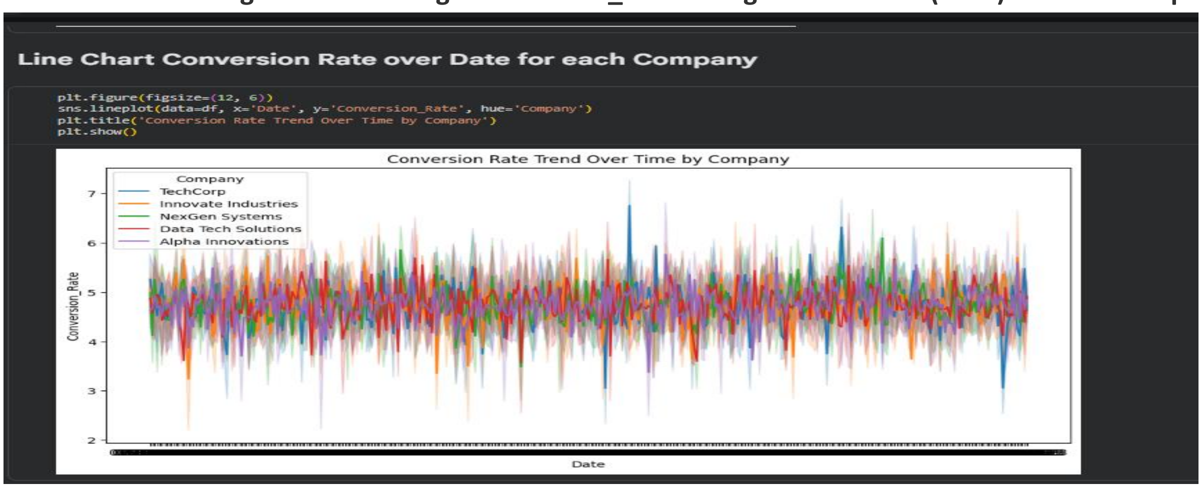
HISTOGRAM CHART (Campaign Duration Distribution)

A histogram showing the distribution of campaign lengths (Duration in days).



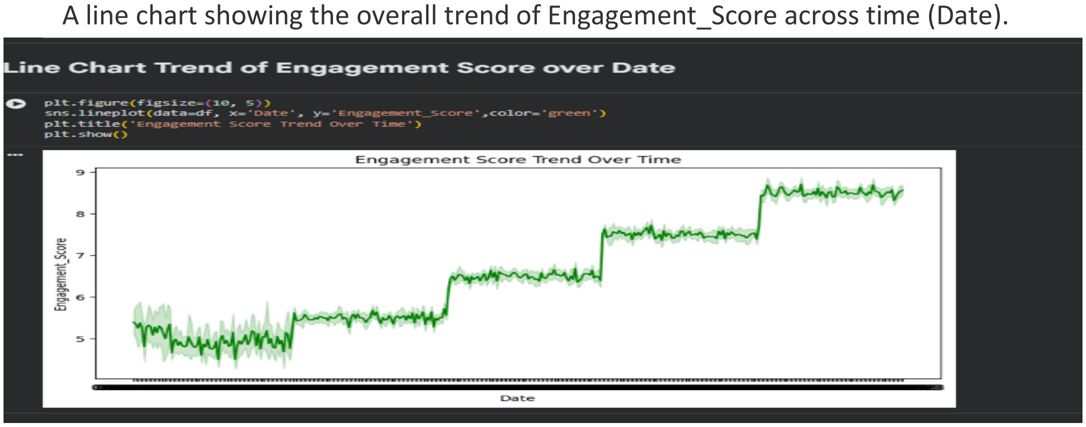
LINE CHART (Company Conversion Rate Over Time)

A line chart tracking how the average Conversion_Rate changed over dates (Date) for each company.



LINE CHART (Engagement Score Trend Over Time)

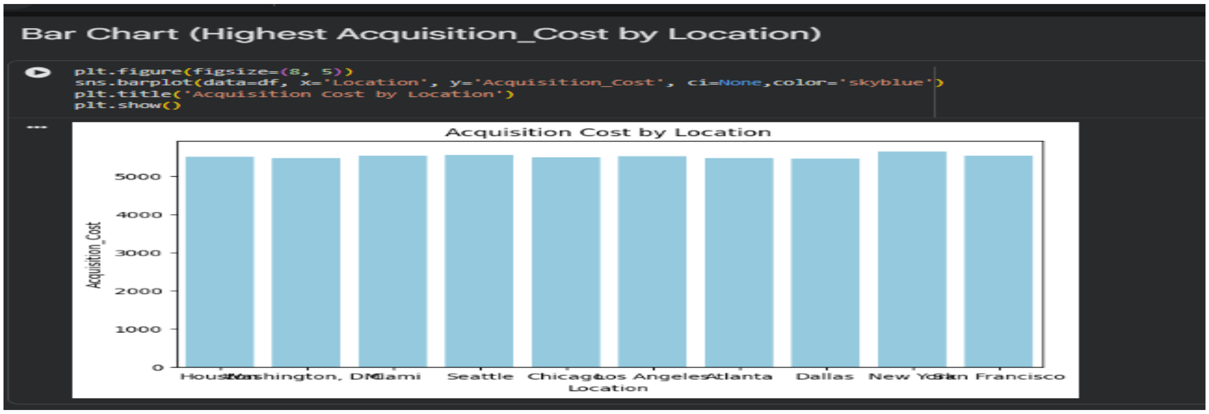
A line chart showing the overall trend of Engagement_Score across time (Date).



Temporal & Geographical Trends

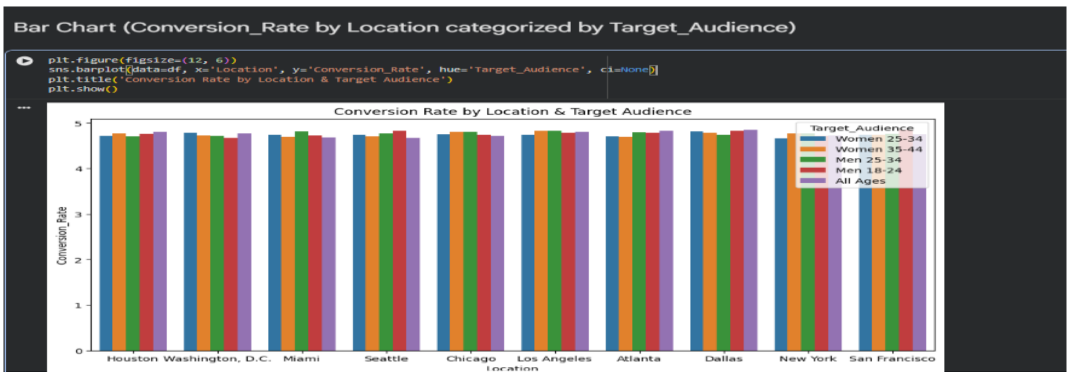
BAR CHART (Campaign Duration Distribution)

Evaluates optimal campaign run lengths (in days) to prevent audience fatigue.



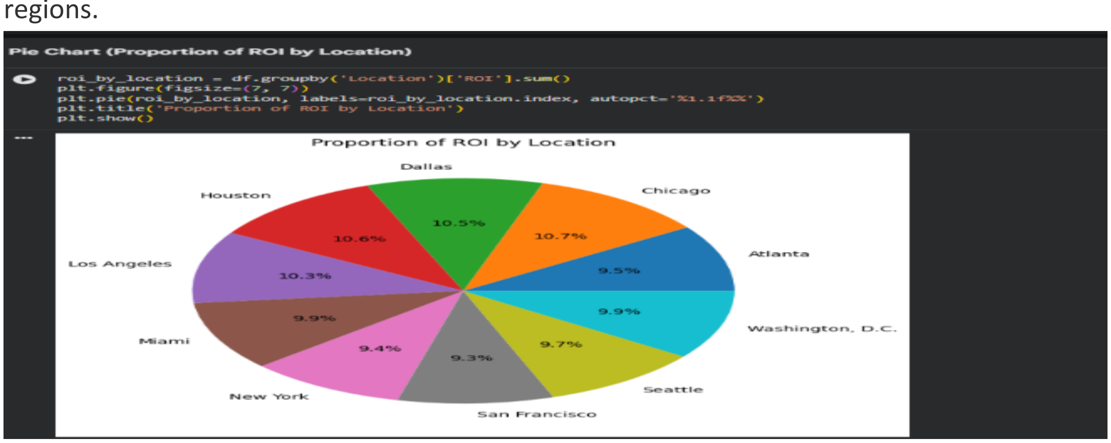
BAR CHART (Time-Series Conversion Dynamics)

Monitors conversion rate fluctuations over time by brand to detect seasonal trends.



PIE CHART (Geographical Cost & ROI Profiling)

Maps acquisition expenses and total profitability across operating locations to prioritize high-yield regions.



Conclusion & Strategic Recommendations

Key Conclusions

ROI & Channel Optimization:

The analysis proves that higher acquisition costs do not automatically guarantee higher ROI Marketing budgets should be reallocated toward channels delivering consistently high Return on Investment rather than high expense platforms.

Target Demographic Alignment:

Conversion efficiency varies significantly across localized languages and target customer segments. Tailoring campaign content to regional language preferences yields noticeably higher conversion rates.

Engagement-to-Conversion Impact:

Higher engagement scores show a positive correlation with conversion success, confirming that interactive and highly engaging campaign formats drive meaningful bottom-line outcomes.

Actionable Recommendations

- **1. Budget Reallocation:** Shift capital away from underperforming channels toward high-ROI combinations identified in the channel effectiveness analysis.
- **2. Personalized Localization:** Expand multi-lingual campaign variations for top-performing customer segments in high-converting regions.
- **3. Optimized Duration:** Structure future campaigns according to optimal duration ranges to maintain maximum user engagement without overspending.